VAERS-Serious: A Medical Text–Tabular Dataset Where Target-Aware Text Representations Pay Off

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Abstract

MulTaBench accepts a text–tabular dataset only if joining text and table beats each alone (*Joint Signal*) and if tuning the text encoder to the target helps further (*TAR gain*), on at least three of five learners; the TAR check is where most candidates fail. We present **VAERS-Serious**, the benchmark's first medical text–tabular dataset: 18,859 U.S. vaccine adverse-event reports, where the task is to predict a serious outcome from twelve structured fields and two short text fields. With the official MulTaBench code, unchanged, it passes both checks on **all five learners and all 25 learner–fold runs**. Its median TAR gain of +0.020 is twice the benchmark median (+0.010) and **ranks third of 21** text datasets; its Joint gain is +0.031 (benchmark +0.023). Both gains hold on folds never used for any design decision (10/10 runs positive). The key design insight is that target-aware text pays off only when the table cannot shortcut the label: matching controls on vaccine, year, age and sex removed that shortcut and raised the TAR gain from at most +0.010 to +0.019–+0.035. Narratives are cut before any mention of care or outcome, and the dataset is rebuilt byte for byte from the raw public files by a single script.[1]

1 Introduction

Many prediction problems mix a table with text: a product listing with its description, a job post with its salary fields, a medical report with its narrative. Tabular foundation models such as TabPFN [7, 9] and strong gradient-boosting baselines [11, 15] handle the structured part well, but they receive text as fixed embeddings from a general-purpose encoder. Recent work shows that tuning those embeddings to the prediction target, *target-aware representations* (TAR), can matter a great deal [1, 2].

MulTaBench [1] turns this into a benchmark-curation rule. Every candidate is evaluated in four conditions: Structured only, Text only, Joint with frozen text embeddings, and Joint with TAR embeddings. A dataset is accepted if, on at least three of five learners, Joint frozen beats both single modalities and Joint TAR beats Joint frozen. By our reconstruction of its published results, only 23 of the 56 text–tabular candidates collected from earlier benchmarks pass (20 of them form its text track), and the most common failure is the TAR check. A good dataset therefore needs a label that truly depends on both modalities, and text whose meaning a generic encoder only partly captures.

We contribute such a dataset from a domain that the text side of the benchmark does not cover: pharmacovigilance. Our contributions are: (i) **VAERS-Serious**, built from the public Vaccine Adverse Event Reporting System with a leakage-controlled and byte-for-byte reproducible pipeline; (ii) an official evaluation in which it passes both checks on 5/5 learners, with a TAR gain that ranks third of 21 and holds on folds held out from all design decisions; (iii) a design insight, supported by version ablations and a permutation test: matched controls remove the table's shortcut and force the model to read the text, which is what makes target-aware tuning pay off.

[1] Code, data and results: https://github.com/DimaBsharaa/text-tabular-research-vaers

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Table 1: From raw reports to the final dataset. Each step removes a specific shortcut or source of noise.

Step	Reports
Unique reports in the three yearly files	208,309
Drop manufacturer-submitted reports (templated; 5.6% vs. 17.0% serious)	131,372
Onset 0–42 days (or unknown); drop infection and vaccine-failure reports	112,726
Merge duplicate reports of the same event	111,127
At least one reaction term after removing outcome and care terms	75,925
Non-empty narrative excerpt (7.9% serious)	63,564
Cut care text from excerpts; drop post-event background	62,754
Matched controls: all serious + up to 3 non-serious per serious report	18,863
Final care cleanup: 503 excerpts shortened, 4 reports removed	**18,859** (26.0% serious)

2 Related work

Text–tabular benchmarks. The Multimodal AutoML benchmark [16] collected 18 tabular datasets with text fields; Grinsztajn et al. [6] studied when language-model embeddings help on tables with strings; CARTE [12] gathered tables dominated by short strings and high-cardinality entries; and TextTabBench [14] focused on longer text fields. MulTaBench [1] pooled these candidates and kept only those that show both Joint Signal and a TAR gain, yielding 20 text–tabular and 20 image–tabular datasets. We follow its curation protocol exactly.

Models. The benchmark's learners are LightGBM [11], CatBoost [15], the MLP ensemble TabM [5], and the tabular foundation models TabPFN-v2 [9] and TabPFN-2.5 [7]. Text is embedded with E5-small-v2 [19] and reduced with PCA [13]; the TAR condition fine-tunes the encoder's last layers with LoRA [10]. Approaches that model text and table jointly, such as TabSTAR [2] or LLM serialisation as in TabLLM [8], are natural future users of the dataset.

Learning from VAERS. VAERS is the U.S. passive surveillance system for vaccine safety, co-managed by the CDC and FDA [17]. Its free-text narratives have been mined to classify reports of specific conditions [3] and to extract adverse events with deep learning [4]. To our knowledge, no prior benchmark pairs the report narrative with its structured fields to predict seriousness.

3 Dataset and curation pipeline

Choosing a source. Our first candidate was tennis: predicting the match winner from rankings and form plus the players' earlier press-conference interviews (3,627 matches). In CPU diagnostics with TF-IDF and logistic regression the text never helped (joint 0.761–0.766 vs. structured 0.768 AUROC; Table 7), interview publication times could not be verified, and the sources restrict redistribution, so we dropped it. The lesson was that the label itself must need both modalities. A VAERS report fits: the structured fields say *who* was vaccinated, *with what* and *when*, while only the narrative says *what happened*.

Source and task. We use the official VAERS yearly files for 2019, 2023 and 2024 [18]: 2019 is a pre-COVID year with the usual mix of vaccines, and 2023–2024 are recent years after the 2021–2022 COVID-19 reporting surge, which would otherwise dominate the data. The target `serious` is 1 if the report's outcome flags record death, a life-threatening event, hospitalisation, prolonged hospitalisation, disability or a birth defect. The flags build the label only; they are never inputs. The task is retrospective classification of a filed report, not a causal claim about vaccines.

Pipeline. Table 1 lists the steps; each was introduced after a concrete failure in an earlier version. Manufacturer reports were dropped because they are templated and let a model learn *who reported* rather than *what happened*. A first version scored 0.88 AUROC from the onset delay alone, because breakthrough COVID-19 infections had to be reported mainly when severe; restricting onset to 0–42 days and removing infection reports fixed it. The full narrative names the outcome directly ("admitted", "died"), and 77% of serious reports involve a hospital stay, so we keep only the **opening** of the narrative, at most three sentences, cut before the first sentence about care, testing, treatment,

Table 2: Official results (AUROC, mean over 5 folds). Joint gain = Joint frozen − max(Structured, Text); TAR gain = Joint TAR − Joint frozen. Both gains are positive for every learner, so both curation checks pass on 5/5.

Learner	Structured	Text	Joint frozen	Joint TAR	Joint gain	TAR gain
LightGBM [11]	0.721	0.778	0.809	0.839	+0.031	+0.030
CatBoost [15]	0.739	0.787	0.823	0.843	+0.036	+0.020
TabM [5]	0.725	0.799	0.832	0.847	+0.033	+0.015
TabPFN-v2 [9]	0.735	0.795	0.821	0.843	+0.027	+0.022
TabPFN-2.5 [7]	0.740	0.805	0.835	0.851	+0.029	+0.016
Median	0.735	0.795	0.823	0.843	+0.031	+0.020

outcome, chart codes or vital signs. A final, label-blind cleanup cut 503 excerpts that still named treatments ("gave Tylenol", "IVIG"); a ledger stores the exact text removed from each report and the build checks every edit. Nothing is rewritten or generated.

Matched controls. Serious reports are 7.9% of the eligible pool, and before matching the vaccine type alone separated the classes (3% serious for shingles vs. 17% for COVID-19). We keep every serious report and, within each stratum of vaccine type, report year, age band and sex, add up to three non-serious reports per serious report, chosen by a fixed hash of the report ID. Matching uses only the label and these four fields, never text. The result is a frequency-matched case–control sample with 26% positives, in which the table cannot win through demographics and vaccine type alone.

The final dataset. VAERS-Serious has 18,859 reports, 4,900 of them serious (26.0%). It has 12 structured columns (age, sex, state, onset delay, five vaccine fields, number of vaccines, facility type, report year) and two text columns: `patient_background` (history, allergies, medications and illness at vaccination, merged; "none reported" in 33% of rows) and `narrative_onset` (the excerpt; median 20 words, 18,336 distinct values). Missing categorical values are kept as their own category and missing numbers are left empty for the benchmark to impute; 1,599 duplicate reports were merged before sampling. Our leakage checks find no outcome, care, chart or procedure terms in the excerpts, excerpt length alone gives AUROC 0.47 (chance level), and none of the 40 words most associated with serious reports is a care or outcome word: they name the kind of reaction (stroke, hypoxia, seizure, Guillain–Barré), while non-serious reports are mild and local (rash, redness, sore arm). The script `build_v11b.py` rebuilds the file from the three raw ZIPs in about three minutes and asserts its SHA-256.

4 Results

Setup. We ran the official MulTaBench `benchmark.py` at commit `c955466` (the `master` branch as of 26 July 2026), unchanged; the only addition is the dataset registration (an enum entry, a file mapping and a curation recipe). Text is embedded with E5-small-v2 and reduced to 30 PCA components per text column. In the TAR condition the last three encoder layers are fine-tuned with LoRA on the target, as a preprocessing step shared by all learners. The five learners use default configurations. Each of the five folds holds out about 1,885 reports for testing, training is capped at 10,000 examples, and the metric is AUROC. All 100 runs (5 learners $\times$ 4 conditions $\times$ 5 folds) took about eight hours on one A100 GPU.

Both checks pass on every learner. Table 2 and Figure 1a show the official scores. Text alone is stronger than the table (median 0.795 vs. 0.735), joining them adds +0.031, and TAR adds a further +0.020. Both gains are positive in all 25 learner–fold runs (Table 3); the smallest TAR gain is +0.0065. With about 1,885 test reports per fold, a single AUROC has a standard error near 0.01, so the conclusion rests on the consistency over 25 runs.

Held-out folds and a permutation test. Every dataset version was screened on folds 0–2 only (mostly with LightGBM and CatBoost). Folds 3–4 were first seen in the official run and stay positive: mean Joint gain +0.036 and TAR gain +0.016 on folds 3–4, against +0.028 and +0.024 on folds 0–2, with 10/10 and 15/15 runs positive. On the screened learners and folds, the official run reproduces

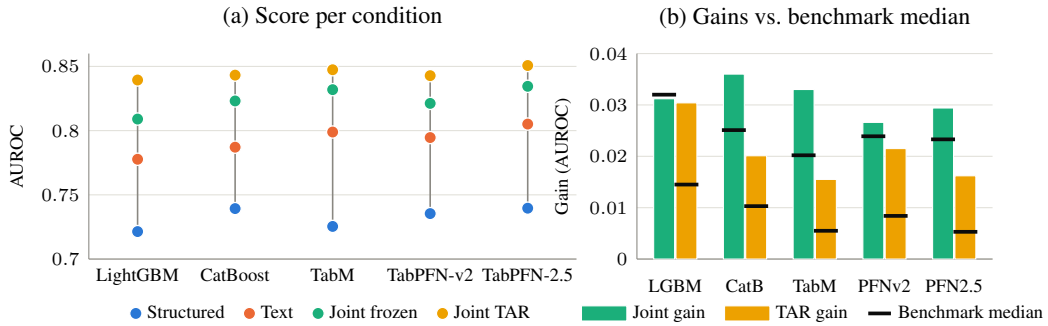


Figure 1: (a) For every learner, joining text and table beats both alone, and target-aware tuning adds more. (b) Our gains (bars) against the per-learner median of the 20 accepted MulTaBench text datasets (black ticks): TAR gain is above the median for all five learners, Joint gain for four of five (LightGBM is just below).

the development screen exactly. To check that the modest structured score is real signal, we trained LightGBM on the 12 structured columns with real and with shuffled labels: the real labels give 0.730 (95% bootstrap CI 0.706–0.755), while 30 shuffles give 0.486 on average and at most 0.514.

What raised the TAR gain. Table 4 tracks the dataset versions. Before matching, the structured columns reached 0.81 and the TAR gain was only +0.006–+0.010. Matched controls removed the table's vaccine-type shortcut (structured 0.74) and raised the TAR gain to +0.021, since the model now has to read what happened. The final cleanup then produced v11b. A control version with v11b's exact rows but the *uncleaned* text shows a larger TAR gain (+0.035 vs. +0.029) but a smaller Joint gain (+0.026 vs. +0.031): its Joint TAR score is the same (0.847 vs. 0.846) and only Joint frozen is lower. The total gain over the best single modality is identical (+0.060 in both), so removing the leaked care text cost nothing in total gain, which shows that the total gain does not come from those words. Split noise reaches about 0.013, so the robust effect is the jump from at most +0.010 without matching to +0.019–+0.035 with it.

Comparison with the benchmark. Against the 20 accepted MulTaBench text datasets, using their released per-learner results, our median gains are +0.031 vs. +0.023 (Joint) and +0.020 vs. +0.010 (TAR). The TAR gain is above the per-learner median for all five learners and the Joint gain for four of five (Figure 1b; Table 5). By median TAR gain the dataset ranks third of 21, after Jigsaw Toxicity and Video Games Sales; note that the benchmark mixes classification (AUROC) and regression (R^2) tasks.

5 Conclusion

VAERS-Serious is a binary text–tabular dataset in which the label needs both modalities and in which a target-tuned text encoder helps beyond frozen embeddings. Evaluated with the official MulTaBench code, it passes both curation checks on all five learners and all 25 runs, with gains above the benchmark median. Its construction is documented step by step, and the dataset is rebuilt exactly from the raw public files.

Limitations. VAERS reports are unverified and self-selected; the label is what the reporter recorded, not a clinical judgement, and nothing here concerns whether vaccines cause these events. The 26% positive rate comes from the case–control design (about 8% in the eligible pool): AUROC is unaffected, but predicted probabilities do not transfer to the full population. We evaluated one encoder size with default settings.

Future work. Models that train text and table jointly [2] may profit most, since fine-tuning already helps here; a temporal split (test on 2024), more years and finer targets (hospitalisation vs. death) are natural extensions.

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A Result tables

Table 3: Official results per fold (AUROC). Both gains are positive in all 25 learner–fold runs.

Learner	Fold	Structured	Text	Joint frozen	Joint TAR	Joint gain	TAR gain
LightGBM	0	0.699	0.794	0.812	0.850	+0.018	+0.038
	1	0.717	0.765	0.800	0.824	+0.035	+0.025
	2	0.723	0.788	0.818	0.855	+0.031	+0.037
	3	0.735	0.765	0.803	0.838	+0.038	+0.035
	4	0.731	0.777	0.812	0.830	+0.035	+0.018
CatBoost	0	0.723	0.804	0.826	0.860	+0.022	+0.033
	1	0.734	0.771	0.816	0.823	+0.045	+0.007
	2	0.738	0.792	0.826	0.861	+0.035	+0.035
	3	0.752	0.782	0.819	0.837	+0.037	+0.019
	4	0.749	0.787	0.828	0.835	+0.041	+0.006
TabM	0	0.705	0.817	0.828	0.859	+0.011	+0.031
	1	0.729	0.784	0.822	0.828	+0.038	+0.007
	2	0.730	0.801	0.846	0.862	+0.045	+0.016
	3	0.735	0.800	0.834	0.849	+0.034	+0.015
	4	0.729	0.793	0.829	0.838	+0.036	+0.008
TabPFN-v2	0	0.721	0.816	0.828	0.860	+0.013	+0.032
	1	0.735	0.781	0.806	0.821	+0.025	+0.015
	2	0.740	0.801	0.826	0.854	+0.025	+0.027
	3	0.744	0.784	0.823	0.841	+0.038	+0.018
	4	0.738	0.791	0.823	0.838	+0.032	+0.015
TabPFN-2.5	0	0.727	0.822	0.846	0.867	+0.024	+0.021
	1	0.734	0.789	0.822	0.828	+0.032	+0.007
	2	0.741	0.813	0.839	0.863	+0.026	+0.024
	3	0.752	0.798	0.834	0.850	+0.036	+0.016
	4	0.745	0.804	0.832	0.845	+0.028	+0.013

Table 4: Development screens of the dataset versions: LightGBM and CatBoost, folds 0–2, mean AUROC (paper-version MulTaBench code). v9_control: v10's rows with v9's text. v11a_control: v11b's rows with the uncleaned v11a text. v11b_context: the two text columns merged. v11c: manufacturer added to the matching strata. v11b was chosen as the cleanest text.

Version	Structured	Text	Joint frozen	Joint TAR	Joint gain	TAR gain
v9 (25% serious)	0.801	0.837	0.864	0.875	+0.028	+0.010
v9_control	0.814	0.838	0.867	0.874	+0.030	+0.007
v10 (care boundary)	0.814	0.833	0.867	0.873	+0.034	+0.006
v11a (matched controls)	0.739	0.775	0.813	0.834	+0.038	+0.021
v11a_control	0.723	0.786	0.812	0.847	+0.026	+0.035
v11b (final)	0.723	0.786	0.816	0.846	+0.031	+0.029
v11b_context	0.723	0.774	0.805	0.832	+0.032	+0.027
v11c	0.732	0.788	0.815	0.833	+0.027	+0.019

Table 5: Gains per learner: ours vs. the 20 accepted MulTaBench text datasets (median, minimum, maximum over datasets).

	Joint gain				TAR gain			
Learner	Ours	Median	Min	Max	Ours	Median	Min	Max
LightGBM	0.0312	0.0320	0.0004	0.1612	0.0304	0.0145	−0.0002	0.1211
CatBoost	0.0360	0.0251	0.0012	0.1774	0.0201	0.0103	−0.0012	0.1029
TabM	0.0330	0.0202	0.0015	0.1800	0.0155	0.0055	−0.0164	0.0969
TabPFN-v2	0.0266	0.0239	0.0027	0.1673	0.0215	0.0084	0.0009	0.1091
TabPFN-2.5	0.0294	0.0233	0.0027	0.1680	0.0162	0.0053	−0.0155	0.0984

Table 6: Median gains over the five learners: VAERS-Serious and the 20 accepted MulTaBench text datasets, sorted by TAR gain (AUROC for classification, R^2 for regression).

#	Dataset	Joint	TAR	#	Dataset	Joint	TAR
1	Jigsaw Toxicity	0.007	0.103	12	Michelin Guide	0.039	0.010
2	Video Games Sales	0.086	0.038	13	SciMagojr Impact	0.105	0.009
3	**VAERS-Serious**	**0.031**	**0.020**	14	Wine Review	0.002	0.007
4	Rotten Tomatoes	0.017	0.018	15	Book Price	0.017	0.007
5	Baby Products	0.024	0.016	16	Women's Clothing	0.016	0.005
6	Product Sentiment	0.081	0.014	17	US Accidents	0.008	0.004
7	Zomato Restaurants	0.023	0.013	18	Spotify Genres	0.036	0.004
8	Kickstarter	0.032	0.013	19	Fake Job Postings	0.010	0.004
9	Mercari Marketplace	0.038	0.013	20	Data Scientist Salary	0.003	0.002
10	Vancouver Salaries	0.168	0.012	21	Montgomery Salaries	0.022	0.002
11	Book Readability	0.023	0.010				

Table 7: The dropped tennis candidate: CPU diagnostics (TF-IDF + logistic regression), development split, AUROC. Adding the text never helped.

Construction	Matches	Structured	Text	Joint	Joint gain
A: earlier interviews, selected policy	3,627	0.768	0.557	0.761	−0.007
A: earlier interviews, best of six	3,627	0.768	0.604	0.766	−0.002
B: serve-placement descriptions	415	0.813	0.603	0.799	−0.014
C: match context written as text	415	0.833	0.458	0.821	−0.012